

Ann MacLarnon

*School of Life Sciences, Roehampton
Institute London, West Hill, London
SW15 3SN, U.K.*

Received 24 June 1994

Revision received 8 September
1995 and accepted 14 September
1995

Keywords: Spinal cord size, brain
size, allometry, primates.

The scaling of gross dimensions of the spinal cord in primates and other species

Previous work on quantitative aspects of the evolution of the spinal cord suggests either that the cord has undergone considerable evolutionary changes, or that it is invariable and somatic. Variation in gross dimensions of the spinal cord is investigated here, mostly in mammals, including new data on 12 primate species, plus some bird and amphibian species. Allometric analyses demonstrate that spinal cord size varies little relative to body size in mammals, including primates. Some gross locomotor differences, but not others, are associated with small differences in relative cord dimensions. Birds, on average, have slightly smaller spinal cords than mammals relative to body size; amphibians may have much smaller spinal cords. On the basis of present evidence, spinal cord weight in mammals scales to body weight with an exponent of 0.69. This is significantly different from the scaling exponent of 0.75 for brain weight to body weight. Therefore, metabolic explanations developed to explain the scaling of brain size are not applicable to the spinal cord. However, the scaling exponent calculated for spinal cord weight is compatible with its scaling to body weight as a surface area to a volume, which may be important for functional interpretation of the scaling of somatic innervation. Simple brain:cord ratios should not be used as measures of relative brain size or intelligence, as the two parts of the central nervous system scale to body size with different exponents, and, at least between classes, because relative spinal cord size also varies. Analysing brain weight relative to spinal cord weight allometrically gives a functionally meaningful assessment of brain size relative to the level of neural communication between the brain and body. Mammals have both large relative brains and cords compared with other classes, which may be required for the sophisticated control of their mode of locomotion. Some species, including the haplorhine primates, have expanded their brain sizes beyond this common level for increased cognitive capacity.

© 1996 Academic Press Limited

Journal of Human Evolution (1996) **30**, 71–87

Introduction

Variation in brain size between living species and through evolutionary time has been the subject of considerable study for a century or more, and there has been a recent upsurge of interest since the publication of Jerison's ground-breaking book (Jerison, 1973). However, little attention has been paid to size variation of the other part of the central nervous system (CNS), the spinal cord. Among those who have been interested in size variation of the cord, two very different schools of thought can be identified. Some researchers, largely those interested in the gross dimensions of the cord, consider the spinal cord is merely somatic, and hence that its size is constant relative to total body size in different species (e.g. Krompecher & Lipák, 1966). Others, usually those more interested in details of particular cord structures, hold that the sizes of these structures, and, therefore, total cord size, are highly variable between species (e.g. Ariëns Kappers *et al.*, 1936). Such workers have rarely used allometric methods to separate adequately variation in cord size associated with overall body size from any residual variation.

However, the few relevant allometric studies provide some support for the idea that spinal cord size has undergone at least some evolutionary change. For example, Towe's (1973) study of variation in the size of the mammalian corticospinal tract showed that the relative size of this cord tract at least, varies among mammals. MacLarnon (in press) has shown that the thoracic spinal cord has increased in girth from Eocene primates to extant forms.



In this study, the scaling of gross dimensions of the spinal cord in primates and some other species is analysed, including new data for primate species. Allometric analyses are carried out to determine whether cord size is an invariable function of overall body size, or whether it varies as a result of evolutionary change, and if so, to what extent.

Possible explanations for the scaling relationships between measures of spinal cord size to body size are also considered. It has been suggested (e.g. Fox & Wilczynski, 1986) that the cross-sectional area of the spinal cord, taken as a measure of the number of nerve fibres connecting the spinal cord to the brain, is likely to scale to body size as a surface area to a volume, because its function is to carry information to and from the body surface area and the brain. On a different basis, metabolic explanations developed to account for the 0.75 scaling exponent for brain weight to body weight (Martin, 1981, 1982; Armstrong, 1982) might be expected to extend to the other part of the CNS, the spinal cord.

The use of spinal cord size as a baseline in studies of relative brain size, either as a measure of body size (e.g. Krompecher & Lipák, 1966), or as a measure of the level of nervous inputs and outputs between the brain and body (Passingham, 1975, 1978) is also investigated.

Materials and Methods

Spinal cord weights were collected for a sample of 16 primate species (including both sexes for three species), and seven non-primate mammal species (including two very different sized subspecies for two species, and both sexes for four species or subspecies). The total mammalian sample was 34, treating males and females and different subspecies as separate data points. In addition, spinal cord weights were collected for five bird species and one amphibian (Table 1).

Spinal cord lengths were collected for a total sample of nine primate species and one non-primate mammal species (Table 1).

Both the samples of spinal cord weights and of spinal cord lengths comprise new data collected here as well as data gathered from the literature (Table 1). A series of 12 non-human primate species, one adult specimen per species, was dissected for measurement of spinal cord weight and length. In nine specimens the cord was preserved using 10% formol-saline in order to minimize shrinkage while adequately preserving the tissue. [Evidence shows that 10% formol-saline causes negligible size changes to cord tissue (Cammarmeyer, 1956).] These specimens are referred to as the FS (formol-saline) subset. The remaining three specimens were preserved similarly but not identically to the FS specimens. They were initially frozen or perfused, apparently unsuccessfully, with a different preservative before preservation in 10% formol-saline (for further details see MacLarnon, 1987). The analyses were performed on the FS subset alone, in case these slight differences affected the cord dimensions, as well as on more inclusive samples.

The whole vertebral column containing the preserved spinal cord was dissected out. The length of each vertebral segment was measured along the ventral surface of the vertebral column, in the sagittal plane. Vertebral segment length at dissection is defined as the ventral length of a vertebral centrum [vertebral measurement number 1 (Martin, 1928)], plus half the length of each adjacent intervertebral disc. The vertebrae were then individually separated, segmenting the cord transversely at the midpoint of each intervertebral disc. The spinal cord segments were removed, stripped of the spinal nerves, the dorsal ganglia and the dura mater, and then weighed.

Table 1 Data

Species	Sex	Body weight (g)	Spinal cord weight (mg)	Brain weight (mg)	Spinal cord length (mm) (Dissection/ Σ DCL)	Refs
Primates						
<i>Daubentonia madagascarensis</i>	U	2800*	—	—	—/223·15	11
<i>Perodicticus potto</i>	U	953·5	—	—	—/171·65	11
<i>Galago crassicaudatus</i>	M	1165*	1924	10 650*	195·5/161·97	1
<i>Callithrix jacchus</i>	F	287*	720	7980*	136·0/117·45	2
<i>Saguinus oedipus</i>	M	325	783	9250*	135·25/112·35	1
<i>Aotus trivirgatus</i>	F	1200	2320	18 110	202·0/180·95	1
<i>Saimiri sciureus</i>	M	805*	1837	24 700*	181·0/167·45	1
<i>Lagothrix lagotricha</i>	M	8160	6733	86 200*	309·25/264·0	1
<i>Presbytis obscurus</i>	F	6360	8287	60 800*	310·5/258·65	1
<i>Miopithecus talapoin</i>	M	1460	2685	37 450	—/192·15	1
<i>Cercopithecus aethiops</i>	M	4878	6300	71 000	—	3
<i>Cercopithecus aethiops</i>	F	3469*	5000	59 000	—	3
<i>Cercocebus torquatus</i>	F	7800	8190	110 500*	320·5/267·0	1
<i>Macaca fascicularis</i>	F	3610*	3790	62 100*	240·3/210·2	2
<i>Papio anubis</i>	M	21 750*	13 231	180 900*	433·0/365·65	2
<i>Papio hamadryas</i>	M	17 950*	9400	165 500*	—	3
<i>Papio hamadryas</i>	F	10 000*	9200	147 300*	—	3
<i>Theropithecus gelada</i>	M	19 410*	12 500	121 000*	—	3
<i>Theropithecus gelada</i>	F	11 690*	10 200	114 500*	—	3
<i>Pan troglodytes</i>	F	34 100*	16 853	374 000	405·0/334·45	1
<i>Homo sapiens</i>	U	60 000*	29 700	1 273 700*	448(♂), 413(♀)/367·1	2, 4, 10
Cetaceans						
<i>Lagenorhynchus obliquidens</i>	M	93 000	33 300	1 157 000	—	5
<i>Lagenorhynchus obliquidens</i>	F	79 000	31 000	1 079 000	—	5
<i>Phocoenoides dalli</i>	F	90 000	29 500	776 500	—	5
<i>Tursiops truncatus</i>	M	86 000	35 000	1 495 000	—	5
<i>Tursiops truncatus</i>	F	70 500	35 300	1 412 000	—	5
Rodents						
<i>Mus musculus</i>	U	19·6	85	430	—	4
<i>Rattus norvegicus</i> (small subspecies)	M	213·5	530	1930	—	4
<i>Rattus norvegicus</i> (small subspecies)	F	211	547	1793	—	4
<i>Rattus norvegicus</i> (large subspecies)	M	273	715	2278	—	6
<i>Rattus norvegicus</i> (large subspecies)	F	258	664	2164	—	6
<i>Cavia porcellus</i>	U	649·4	1100	4000	162·98(♂)/—	7
Lagomorphs						
<i>Oryctolagus cuniculus</i> (small subspecies)	M	1972	3945	9265	—	8
<i>Oryctolagus cuniculus</i> (small subspecies)	F	2480	4537	9755	—	8
<i>Oryctolagus cuniculus</i> (large subspecies)	M	3352	5781	11 711	—	9
<i>Oryctolagus cuniculus</i> (large subspecies)	F	3579	5940	11 557	—	9
Birds						
<i>Adapornis personata</i>	U	40	120	1920	—	4
<i>Passer domesticus</i>	U	27	85	1000	—	4
<i>Hirundo rustica</i>	U	18	80	535	—	4
<i>Hirundo rustica</i>	U	20	75	492	—	4
<i>Columba domestica</i>	U	335	620	2030	—	4
<i>Columba domestica</i>	U	326	575	1790	—	4
<i>Gallus domesticus</i>	F	1340	1554	3525	—	4
<i>Gallus domesticus</i>	M	1800	2175	3680	—	4
Amphibian						
<i>Rana esculenta</i>	U	48	54	117	—	4

References: (all data for a species come from a single source except where indicated).
*Martin & MacLarnon (unpublished data collection), best available weight for specimens of the same sex; 1 Present study (FS subset); 2 Present study (non-FS subset); 3 Hopf & Claussen (1971); 4 Krompecher & Lipák (1966); 5 Ridgway *et al.* (1966); 6 Donaldson & Hatai (1911); 7 Latimer (1950); 8 Latimer & Sawin (1955); 9 Latimer & Sawin (1957); 10 Ravenel (1877); 11 Nayak (1933).

Vertebral measurements were taken for 45 skeletal specimens, representing 38 primate species. This sample included the 12 specimens for which spinal cord data were collected here. For each specimen, the dorsal centrum length [vertebral measurement number 2 (Martin, 1928)] was measured for each vertebra in the sagittal plane, excluding the length of the odontoid peg of the axis.

Spinal cord lengths were calculated in two different ways for the spinal cords dissected here. Dissection-type spinal cord lengths were calculated using the measurements of segment lengths taken at dissection, and DCL-type spinal cord lengths were calculated using the measurements of dorsal centra lengths (DCL) taken on cleaned vertebrae. These spinal cord lengths are the sums of the lengths of the vertebral segments at dissection, or of the cleaned vertebrae, respectively, from the cranial end of the atlas to the midpoint along the length of the vertebra in which the spinal cord ended. The total presacral length of the vertebral column was also calculated for the total sample of 38 primate species as the sum of the dorsal centrum lengths of all their presacral vertebrae.

Additional spinal cord data were collected from published sources (Table 1). Only the weights of fresh cords were included; no studies used weak formol-saline, and other preservatives may not give comparable results. For mammals, only clearly adult specimens were included, according to species' body weights (Martin & MacLarnon, unpublished data collection).

Only human (Donaldson & Davis, 1903) and guinea pig (Latimer, 1950) data were found in a literature search for spinal cord length measurements comparable with the dissection-type lengths measured in the present study. Donaldson and Davis (1903) give adult human measurements from several sources. The data they quote from Ravenel (1877) were used here, as the 11 male and 11 female cords were measured fresh (male mean=448 mm, range 390–480 mm; female mean=413 mm, range 370–460 mm). The other measurements given by Donaldson and Davis (1903) lie within these ranges, as do the figures generally quoted in anatomy textbooks; e.g. Romanes (1972) gives the range of normal human spinal cord lengths as about 430–450 mm.

Several DCL-type spinal cord lengths were calculated from information available in the literature plus vertebral measurements taken in the present study. On average the human adult spinal cord ends at the border between the first (L1) and second (L2) lumbar vertebrae (e.g. Romanes, 1972). Using measurements of a human vertebral column, the human spinal cord length was calculated as the sum of the dorsal centra lengths from the atlas to the caudal end of L1.

Nayak (1933) gives the vertebral position at which the spinal cord ends for two other species for which vertebrae were measured in the present study. The cord ends between the third and fourth lumbar vertebrae in *Perodicticus potto* and within the last lumbar vertebra in *Daubentonia madagascarensis*. Spinal cord lengths for *Perodicticus* and *Daubentonia* were calculated in a similar way to that for *Homo sapiens*.

Brain weights were also measured for the specimens dissected in this study, wherever the brain was preserved well enough ($n=3$). For the remaining specimens, brain weight data from other sources for the appropriate species and sex are used (Martin & MacLarnon, unpublished data collection) (Table 1). Brain weight data were published with all the spinal cord weights collected from the literature.

The body weights used in analyses are the actual body weights of the specimens from which spinal cord data were collected wherever these were available, and, for primates, whenever they fell within the range for wild-caught specimens of the species of the same sex (Martin &

MacLarnon, unpublished data collection) ($n=7$). For the remaining species, the body weights used in analyses are species' averages for the appropriate sex (if known), using data from wild-caught specimens wherever possible (Martin & MacLarnon, unpublished data collection) (Table 1).

The best-fit lines for scaling relationships were calculated as principal major axes. There is continuing debate about the most appropriate line-fitting technique for describing allometric relationships and calculating residual variation from these relationships (e.g. Aiello, 1992; Pagel & Harvey, 1988; Martin & Barbour, 1989; Smith, 1994). However, where correlation coefficients are high there is very little difference between the results of the major methods (Aiello, 1992).

Results

Scaling of spinal cord weight to body weight

The relationship between spinal cord weight and body weight is presented in Figure 1. The correlation coefficient for the mammalian sample is very high (0.99), and the distribution of mammalian points on Figure 1 clusters tightly about the best-fit major axis, indicating that there is little variation in cord weight relative to body weight (relative cord weight) among mammals. It is possible that a larger and more varied sample would show more variation in relative cord weight, but the present sample contains representatives of groups that have very different relative brain sizes, and yet there is no indication of such variation in the relative size of the other part of the CNS, the spinal cord.

The rabbits have the largest relative spinal cord weights of all the mammalian sample species, and some of the cetaceans have the smallest. The primates, including *Homo sapiens*, all fall within the range of deviation of the non-primate mammals from the mammalian line, demonstrating that they do not have distinctive relative spinal cord weights compared with the other mammals. The major axis for the scaling of spinal cord weight to body weight for the FS subset of primate species alone is also very similar to that for all mammals (Table 2), again indicating that the primates are typical mammals in terms of the relationship between spinal cord weight and body weight.

The bird points on Figure 1 all fall below the mammalian line. Most of them are at least as far below the line as the most deviant mammals, and several lie further below the line than any of the mammals. The one amphibian point lies far below the mammalian line. This evidence suggests that birds, on average, have smaller relative spinal cord weights than mammals, though there is some overlap, and that at least one amphibian has a much smaller relative cord weight than both mammals and birds.

Scaling of spinal cord length and vertebral column length to body weight

The relationships between the two measures of spinal cord length and body weight are presented in Figure 2. The correlation coefficients for the distributions of the FS subset are both very high (≥ 0.98), and all the points plotted cluster closely about the lines. For all the species in Figure 2, except *Homo sapiens*, species' points are based on individual specimens. In addition, many of the calculations of spinal cord lengths involved estimations that spinal cords ended at the mid-point of a vertebral segment, or at one end. These factors are limitations on the analysis of variation in relative cord lengths that were possible in this study. Because any errors they introduce will be randomly distributed, however, there is no reason to doubt the main conclusion that there is little variation in relative cord length in primates. The data from

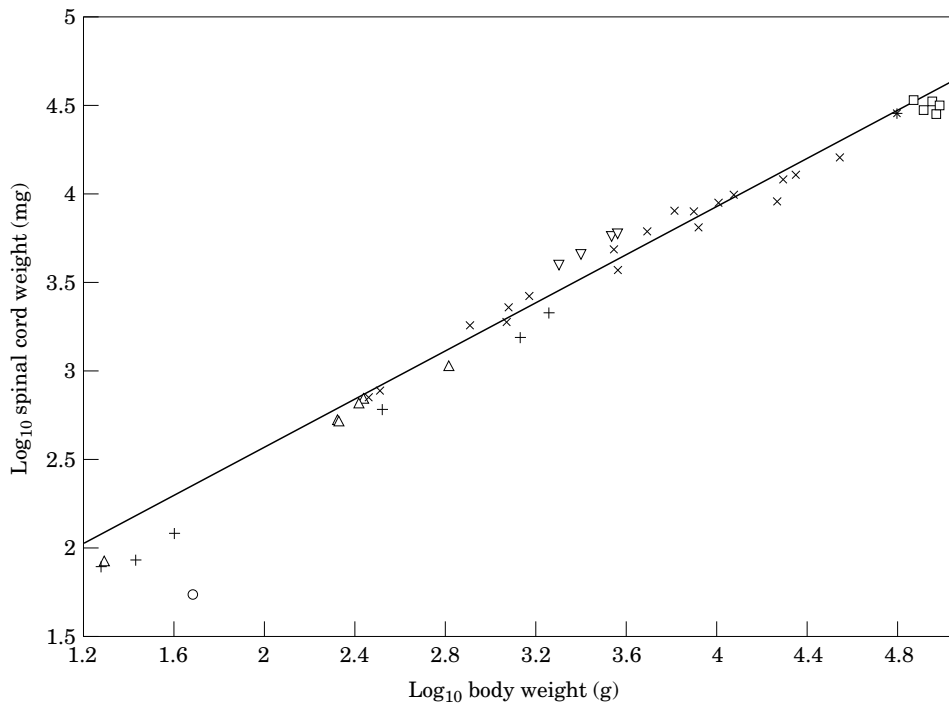


Figure 1. Log of spinal cord weight *vs.* log of body weight for the total available sample of mammals, birds and an amphibian. The line is the major axis for mammals ($y=0.69x+1.19$). *, *Homo sapiens*; x, other primates; □, cetaceans; Δ, rodents; ▽, lagomorphs; +, birds; ○, amphibian.

one non-primate mammal, *Cavia*, also give no indication that the relative spinal cord lengths of primates are any different from those of other mammalian species.

Homo sapiens and the largest of the non-human primates, *Pan*, fall first, second or third furthest below the two lines on Figure 2, suggesting that taillessness might be associated with a slightly reduced relative cord length. The points for *Pan* are based on a single specimen only. However, the human dissection-type cord length is an average for 22 individuals. If the full range for this sample were plotted on Figure 2, the longest cord lengths would only just fall within the range of deviation of the other primate species from the best-fit line. It seems probable, therefore, that the relative spinal cord lengths of tailless primates are a little shorter than those of other species.

In order to compare the scaling of spinal cord length with that of the vertebral column, the major axis for the scaling of the total length of the presacral vertebral column to body weight was calculated for the total primate sample measured, less the hominoids, which tend to have relatively short trunk lengths (Aiello, 1981) (Table 2).

Relationship between brain weight and body weight

The relationship between brain weight and body weight for the total sample collected is presented in Figure 3. The line drawn in is not the best-fit line for the present sample, but a line reflecting the well-established exponent of about 0.75 for the scaling of brain weight to body weight in mammals (e.g. Martin, 1981). The principal major axis for the present sample alone indicates a much higher exponent of 1.02 (Table 2). The reason for the large difference

Table 2 Allometric statistics

<i>x</i> variable	<i>y</i> variable	Sample	<i>n</i>	Major axis slope	95% confidence limits on major axis slope	Major axis intercept	<i>r</i>
Body weight	Spinal cord weight	FS subset	9	0.66	0.60–0.73	1.30	0.99
Body weight	Spinal cord weight	Mammals	34	0.69	0.66–0.72	1.19	0.99
Body weight	Spinal cord length (dissection-type)	FS subset	8*	0.24	0.22–0.26	1.56	0.99
Body weight	Spinal cord length (ΣDCL-type)	FS subset	9	0.22	0.19–0.26	1.54	0.98
Body weight	Presacral vertebral column length	Primates less hominoids	32	0.27	0.24–0.31	1.44	0.93
Body weight	Brain weight	Mammals	34	1.02	0.94–1.11	0.97	0.96
Spinal cord weight	Brain weight	Mammals	34	1.52	1.36–1.71	–0.91	0.95
Spinal cord weight	Brain weight	Non-human primates	18	1.17	1.03–1.34	0.44	0.96
Spinal cord weight	Brain weight	Rodents and lagomorphs	10	0.78	0.75–0.82	1.14	1.00

*Dissection-type measurements were not taken for *Miopithecus talapoin*, hence the smaller FS subset in this case.

between the two exponents is that the present sample of mammal species contains a number of well-recognized grades for relative brain size, and these are not randomly distributed with respect to body weight. This can be seen in Figure 3: rodents and lagomorphs, which include the smallest sample species, lie well below the line in Figure 3, indicating that they have the lowest relative brain weights of the sample; the non-human primates, which fall in the middle of the size range, form a fairly well defined grade of higher relative brain size, all but the one prosimian having larger relative brain weights than the typical mammalian condition; *Homo sapiens* and most of the cetaceans, the largest species, have higher relative brain sizes than all the non-human primates. As a result of this pattern, the empirical line for the scaling of brain weight to body weight based on the present sample alone suffers from the effects of grade confusion (Martin, 1980), and is tipped upwards to give an extremely high slope value.

The position of the bird species in Figure 3, shows that some of the sample birds have relative brain sizes within the range of the mammals, and others have relative brain sizes smaller than all the sample mammals. The one amphibian species has a much smaller relative brain size than all the other species.

Relationship of brain weight to spinal cord weight

The relationship between brain weight and spinal cord weight is presented in Figure 4. No line has been drawn in because it is difficult to determine which subgroup(s) should be used for its calculation on the basis of the relatively small and diverse sample. The slopes of the major axes for all mammals, the non-human primates and rodents+lagomorphs are very different from one another (Table 2). The distribution of points in Figure 4 is similar to that for brain weight *vs.* body weight (Figure 3). The main differences are that on the plot against spinal cord weight (Figure 4), the one amphibian point is not so outstanding from the rest of the species, and none of the bird points are so clearly outside the mammalian range as some are on the plot against body weight (Figure 3).

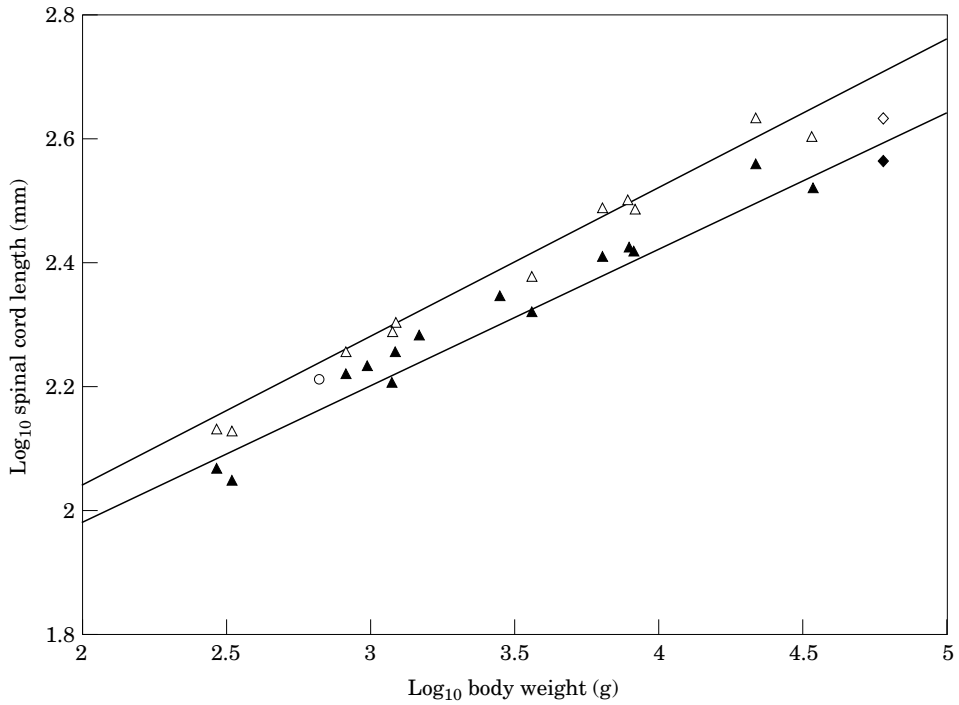


Figure 2. Log of spinal cord length vs. log of body weight for primates plus *Cavia porcellus*. The lines are the major axes for the FS subset (dissection-type spinal cord length: $y=0.24x+1.56$; $\diamond$, *Homo sapiens*; $\triangle$, other primates; $\circ$, *Cavia porcellus*) (DCL-type spinal cord length: $y=0.22x+1.54$; $\blacklozenge$, *Homo sapiens*; $\blacktriangle$, other primates).

Discussion

The results presented here provide evidence relevant to various issues concerning the variation and evolution of spinal cord dimensions.

First, present analyses indicate that gross dimensions of the spinal cord, its total weight and length, are highly correlated with body size in mammals, including primates, suggesting that there is little variation in relative cord dimensions within this class. Although the present samples are quite small, these findings agree with those of Fox & Wilczynski (1986) and Hofman (1982), based on analyses of the cross-sectional area of the spinal cord at its junction with the medulla. They showed that this dimension of the spinal cord is highly correlated with body size in rodents and some other mammals. However, as the results of studies by Dagosto & Terranova (1992) and Aiello & Wood (1994) demonstrate, high correlation coefficients for allometric plots of logarithmically transformed data can disguise important amounts of variation in the relative absolute size of variables. Using a similar measure to these studies, 85% of mammalian species and 95% of primate species in the present sample have spinal cord weights within $\pm 20\%$ of the values expected for typical mammals, using the equation in Table 2. However, species do vary from expected values by as much as $+36\%$ for *Oryctolagus cuniculus* and $+27\%$ for *Presbytis obscurus*. This form of analysis still indicates that variation in relative cord weight is limited among mammals in general, but it shows that some potentially functionally important differences are present. From more limited evidence, it appears that

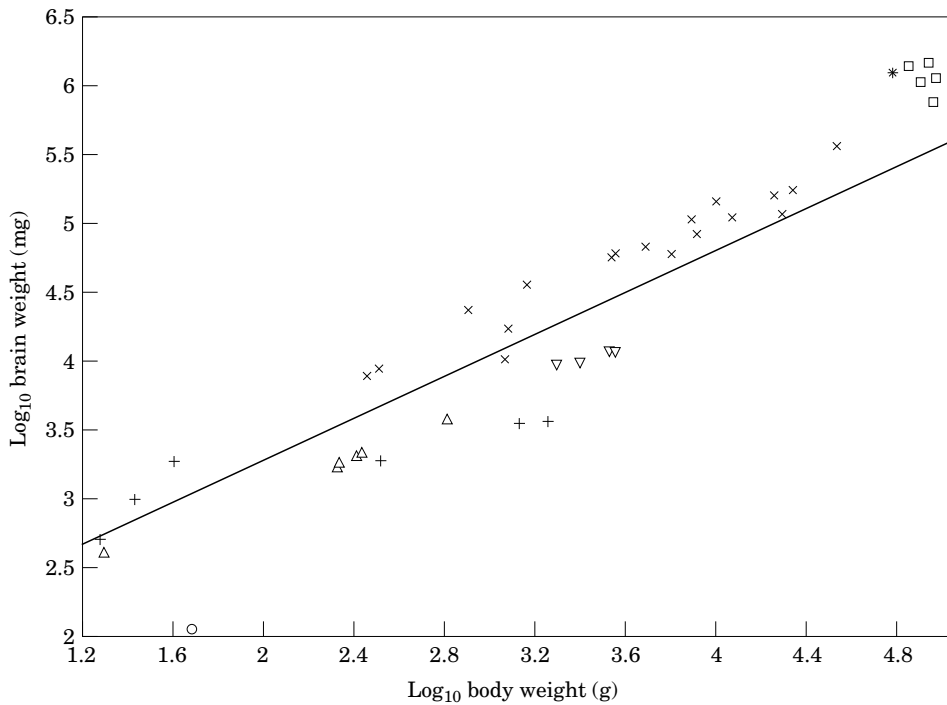


Figure 3. Log of brain weight *vs.* log of body weight for the total sample of mammals, birds and an amphibian. The line is the major axis for mammals, calculated for a sample of 313 species (Martin, 1981) ($y=0.76x+1.77$). Key as for Figure 1.

birds, on average, have smaller relative cord weights than mammals, and data from a single amphibian species suggest that this class may have much smaller relative cord weights than both the others.

This overall picture has some marked differences from that portrayed by many comparative neuroanatomists, particularly those who work on variation in individual cord structures. Their results have commonly suggested that there is considerable variation between the spinal cords of different vertebrate species, in the number and size of tracts present, including among mammals (Ariëns Kappers *et al.*, 1936; Nieuwenhuys, 1964; Sarnat & Netsky, 1981). As examples, the spinothalamic tract has been identified as having increased progressively with evolution to "higher" forms, particularly in the mammals (Mehler, 1969), and the spinocerebellar tract has apparently increased proportionately with increase in the size of the cerebellum (Sarnat & Netsky, 1981). Together, such analyses imply that total gross cord dimensions vary greatly within, as well as between, orders and classes, in general increasing from "lower" to "higher" vertebrates, and "lower" to "higher" mammals.

That the analyses presented here suggest a different interpretation can probably largely be accounted for by the use of allometric methods and separate consideration of major taxa in this study. The previous work frequently involved the use of simple ratios and small samples of species covering wide taxonomic ranges. This did not enable size-related variation and size-free evolutionary changes to be separated clearly. Simple ratios of the size of one part of the cord to another have been used to compare the relative sizes of homologous cord structures in species of different body sizes (e.g. Ariëns Kappers *et al.*, 1936). However, such ratios may

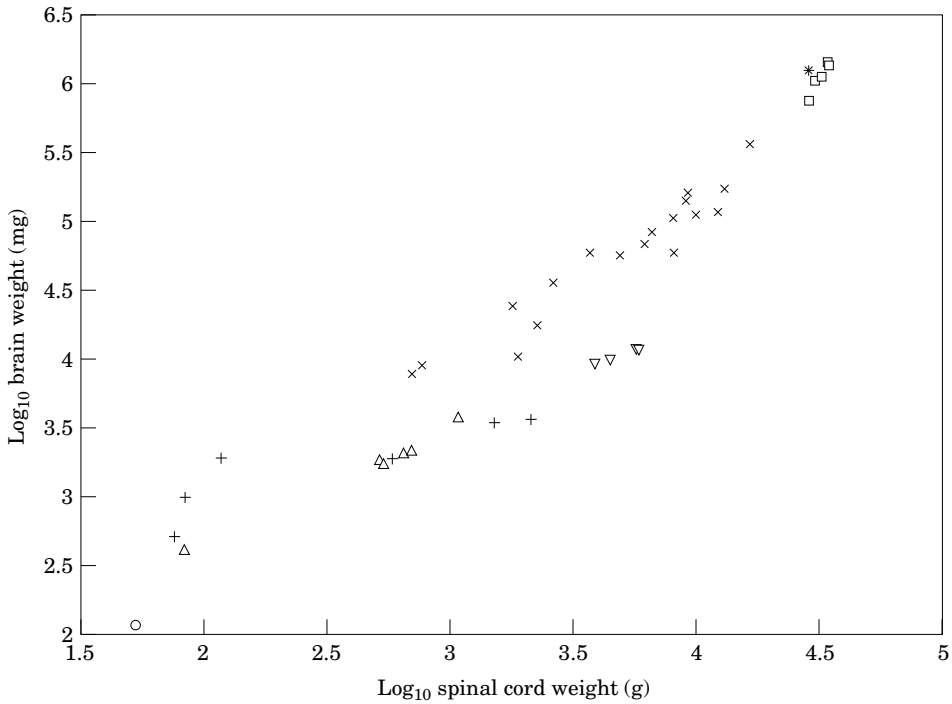


Figure 4. Log of brain weight *vs.* log of spinal cord weight for the total available sample of mammals, birds and an amphibian (see Table 2 for statistics). Key as for Figure 1.

not be “size-free”, that is free from the influence of overall body size, because different parts of the cord may not scale to body size with the same exponent. Hence, their ratios may intrinsically increase, or decrease, with increasing body weight. Small, wide-ranging samples also make it especially difficult to establish whether a difference in a ratio is the result of evolutionary change in the relative size of the numerator feature, or the denominator, or both. Previous studies have usually assumed that it is the numerator feature only that varies.

A few previous studies of individual cord structures have used allometric methods of analysis and more representative samples of species. These have commonly identified only limited variation in the relative sizes of tracts and other features, much more in keeping with present results for the total cord. For example, Towe (1973) showed that the size of the corticospinal tract relative to body size does vary in mammals, but only to a limited extent. Species form two quite clear-cut grades with relatively larger and smaller tracts. MacLarnon (1993) demonstrated that most primates show no significant variation in the amount of spinal grey matter in the thoracic region, once body size is taken into account, though humans have increased the grey matter a little in this region.

Working from a very different perspective, another group of comparative neuroanatomists have essentially made the assumption that spinal cord size directly reflects the size of the body, and, hence, that simple ratios of brain weight to spinal cord weight are “size-free” measures of relative brain size (Latimer, 1950; Warden, 1951; Krompecher & Lipák, 1966; Ridgway *et al.*, 1966; Hopf & Claussen, 1971). This has been justified on the qualitative grounds that it seems functionally reasonable to compare the complex, mental, developing brain with the

simple, somatic, unchanging spinal cord. However, as described above, for such simple ratios to be meaningful, a number of more quantitative requirements must be met. The present results show that between classes, at least, there are significant differences in the size of the spinal cord relative to body size. Therefore, direct comparison of simple brain:cord ratios between, say, birds and mammals, cannot be interpreted solely in terms of variation in relative brain size.

In summary, previous comparative anatomical studies involving quantitative measures of the spinal cord have mostly either indicated that it has undergone considerable evolutionary changes, or that it has been evolutionarily inert. Present analyses, and a few more methodologically rigorous previous studies, suggest an intermediate picture. They demonstrate that although some change has taken place in quantitative features of the cord producing variation among extant species, this has apparently been limited within the mammals, and perhaps other classes, with some more major changes taking place between classes.

Previous studies that have identified quantitative differences between the spinal cords of different species have commonly associated these with differences in locomotor behaviour, either in the general level of locomotor sophistication, or particular locomotor adaptations (e.g. Ariëns Kappers *et al.*, 1936; Sarnat & Netsky, 1981). Although the actual degree of quantitative variation in the cord which is not size-related may be much less than many of these studies have suggested, it remains possible that variation in locomotor behaviour is an important influence on the relative size of the dimensions of the gross spinal cord and its constituent structures. Within the mammals, there is a little evidence for this from analyses presented here. Some cetaceans, which lack hindlimbs, have somewhat small spinal cord weights for their body weights, and rabbits, with their very powerful hindlimbs, have somewhat large relative cord weights, which may reflect their respective reduced and increased requirements for hindlimb innervation. However, within the primates there is little variation in relative spinal cord weight, despite considerable locomotor variation among the sample species, and what variation there is shows no clear relationship with locomotor variation. The terrestrial and arboreal quadrupeds span the full range of primates, having relative cord weights of between -19% and $+27\%$ of expected values for typical mammals of their body sizes. The hindlimb leaper, *Galago*, has a relative cord weight only very slightly below expected size (-5%), and does not have an expanded cord like the rabbits. *Lagothrix*, with its prehensile tail or "fifth limb", shows no sign that this requires an overall increase in innervation, having a relative cord weight 13% below expected. The biped, *Homo*, has a relative cord weight just 3% below expected, and is not at all outstanding. Overall, therefore, locomotor variation may have some influence on relative cord weight in mammals, but this is neither consistently marked nor clear-cut.

Previously, it has also been suggested that relative tail length and usage have considerable effects on relative cord length among vertebrates, including among mammals. For example, Nayak (1933) suggested that relative cord length in prosimians is linked to their tail length. In terms of the vertebra in which the cord ends, it ends most caudally in the long-tailed *Galago senegalensis* and *Daubentonia*, more cranially in the short-tailed *Perodicticus*, and most cranially in the tailless *Loris*. Among monotremes with similar tail lengths, the spinal cord is apparently much shorter in *Echidna*, which has a thin, little-used tail, than in *Ornithorhynchus*, which has a powerful, muscular tail used in swimming (Ariëns Kappers *et al.*, 1936).

Present analyses suggest that within the narrow range of variation in relative cord length among primates, the tailless species, represented by *Pan* and *Homo sapiens*, have the shortest relative spinal cord lengths of the order. However, the one sample primate species with a

relatively short tail, *Perodicticus*, does not have a relatively short spinal cord length, as might have been expected; and the one sample species with a prehensile tail, *Lagothrix*, does not have a particularly long spinal cord relative to body weight, despite the probable extra innervation required for its strong, sensitive tail. Apparently, only complete loss of the tail in primates is associated with variation in the relative length of the spinal cord.

The length of the spinal cord in previous studies was usually described according to the position of the vertebra in which the spinal cord ended, e.g. in the fourth lumbar vertebra. This may not directly reflect relative cord length, which was used in analyses here, because of several possible complicating factors. First, the species being compared might have different numbers of vertebrae in the various vertebral regions and/or in total. Second, though not necessarily independently, species may differ in the relative length of the vertebral column. Third, the length of the vertebral column need not scale to body weight with the same scaling exponent as spinal cord length, and, therefore, there may be a general tendency for the spinal cords of larger or smaller species to end relatively caudally or cranially with respect to the vertebral column. This third possible complicating factor is apparently not involved, at least in primates, as the scaling exponents for presacral vertebral column length and spinal cord length calculated here are virtually identical (0.27 and 0.25/0.26). Nevertheless, for the other reasons given, variation previously noted in the endpoints of the spinal cords of other mammalian species relative to their vertebral columns should not be taken as good evidence for variation in their relative spinal cord lengths without further investigation.

It remains possible that the major locomotor differences between mammals, birds and amphibians may be related to the differences identified between these classes in their relative spinal cord weights, reflecting variation in the amount of brain–cord communication, despite the lack of evidence for a close correlation between locomotion and relative cord size among mammals. However, without larger samples, the evolutionary cause(s) of the differences in total relative cord weight between mammals, birds and amphibians cannot be interpreted with any certainty.

The limited variation in the relative size of the mammalian cord identified in this study is very different from the other part of the CNS, the brain, which varies considerably within the class as a whole, including among the sample species analysed here. Present results also give no sign of correlation between relative cord weight and relative brain weight in primates and other mammals. Even taking the narrower range of relative cord size into account, the distributions of species on the plots of cord weight and brain weight against body weight (Figures 1 and 3) are very different. For example, the lagomorphs have the largest relative cord weights of the whole sample, whereas they have relative brain sizes below that of typical mammals. The primates all have relative cord weights within the range of other mammals, and the haplorhines, to which all the sample primates except one species belong, do not have relatively large cord weights compared with other mammals, unlike their relatively large brain weights. *Homo sapiens* has a relative cord weight very close to the typical size both for primates and for all mammals, even though humans have a much larger relative brain weight than most other species. Therefore, the selection forces operating on the gross dimensions of the spinal cord and brain among mammals must have been very different.

On the basis of the results from the present study, the scaling exponent for spinal cord weight to body weight is about 0.69 in mammals, and about 0.66 for primates alone. The difference between these two best estimates is statistically insignificant, and both values are significantly lower than the established scaling exponent of 0.75 for brain weight to body weight in mammals (e.g. Martin, 1981). On statistical grounds, it could be argued that the

scaling of brain weight and spinal cord weight to body weight should be compared using the same sample of species. If the relationship of brain weight to body weight is analysed for the same mammalian sample as for the analysis of the scaling of spinal cord weight to body weight, the scaling exponent is 1.02. The 95% confidence limits on this value do not overlap with those for the scaling exponent for spinal cord weight calculated for the same sample, again indicating that the two parts of the CNS scale differently. However, analysing the scaling of brain weight to body weight on the basis of this restricted sample is not actually biologically meaningful because of the problems of grade confusion described above. Such problems might also affect the analysis of the scaling of spinal cord weight. The very high correlation coefficient (0.99) for the relationship between spinal cord weight and body weight for the present sample suggests that this is not the case, but it remains possible that a larger sample would give a significantly different result. However, with this proviso, the evidence from this study indicates that spinal cord weight does not scale to the 0.75 power of body weight, unlike brain weight. Therefore, suggested metabolic explanations for the scaling of brain size to body size (Martin, 1981, 1982; Armstrong, 1982) are not applicable to the other part of the CNS.

The differential scaling of spinal cord weight and brain weight to body weight also provides a second reason why simple ratios of brain to cord weight are not suitable as measures of relative brain size or intelligence, as several previous authors have assumed (Latimer, 1950; Warden, 1951; Krompecher & Lipák, 1966; Ridgway *et al.*, 1966; Hopf & Claussen, 1971). Brain:cord ratios will tend intrinsically to increase with increasing body size, because brain weight scales to body weight with a higher exponent than spinal cord weight. Hence, the use of brain:cord ratios would generally overestimate the relative brain sizes of larger-bodied species compared with smaller-bodied forms.

Previous estimates of the scaling of spinal cord size to body weight in mammals by Hofman (1982) and Fox & Wilczynski (1986) have used a measure of the cross-sectional area of the spinal cord at its junction with the medulla. The average scaling exponent for the cross-sectional area of the spinal cord can be estimated for the primate sample in the present study by combining the scaling exponents calculated for the scaling of spinal cord weight (a volumetric measure; 0.66) and of spinal cord length calculated from measurements taken at dissection (0.26). This gives a value for the average scaling exponent for spinal cord area of 0.40 (0.66 - 0.26), which falls within the 95% confidence limits of the values calculated in the previous studies mentioned, which were 0.341 ± 0.155 , and 0.403 ± 0.007 , respectively (Hofman, 1982; Fox & Wilczynski, 1986).

Hofman (1982) and Fox & Wilczynski (1986) were interested in the cross-sectional area of the spinal cord as a potential measure of somatic innervation, for use in the assessment of relative brain size in terms of "extra capacity", beyond that necessary to meet basal requirements (Jerison, 1973). As well as its being functionally appropriate, more quantitative analyses presented here and in these previous studies also support the suitability of spinal cord size for this purpose within mammals, because it is highly correlated with body size in this group. Fox & Wilczynski (1986) point out that the cross-sectional area of the spinal cord might be expected to scale to body weight as a surface area to a volume, i.e. with a scaling exponent of 0.67, following the surface area theory for the scaling of brain size, perhaps because information flows largely to and from body surfaces. However, empirical findings demonstrate that this is not the case, which may, as Fox & Wilczynski (1986) suggest, result from one or more complicating factors confounding the use of the cross-sectional area of the spinal cord as a direct surrogate for the number of nerve fibres communicating between the cord and the brain. For example, the constituent parts of total spinal cord size, such as the numbers of

various cell types and nerve fibre size itself, may not individually scale isometrically with total cord size; and hence overall scaling exponents of spinal cord size may not reflect simply the underlying functional cause of the rate of spinal cord scaling. Similar problems might include organizational change or aspects of network allometry as suggested for brain size scaling by Deacon (1990*a,b*) and Ringo (1991). It would also be better to use a measure of the white matter of the cord alone to reflect numbers of nerve fibres, although, as Fox & Wilczynski (1986) pointed out, their data come from the highest cervical levels of the cord where there is relatively little grey matter.

However, while the cross-sectional area of the spinal cord does not scale to body size as a surface to a volume, the exponents calculated here for primates and for all mammals for the scaling of spinal cord weight to body weight are both insignificantly different from 0.67. The size of the total spinal cord may in fact be a good measure of somatic innervation, taking into account, for example, the length of nerve fibres as well as their cross-sectional area. Functional interpretation of the scaling exponent for the total spinal cord is difficult, because it results from the aggregation of the scaling exponents of many constituent factors. However, the possibility of a surface area to volume relationship between total cord size and body size clearly should not be ruled out.

Although simple brain:cord ratios are not useful measures of relative brain size, as described above, spinal cord size might be justified as a useful baseline for the assessment of relative brain size or intelligence on other grounds, providing allometric methods of analysis are used. If spinal cord size reflects the level of nervous inputs and outputs to and from the brain and body, then, regardless of whether this is directly correlated with body size, it might be meaningful to analyse brain size relative to cord size (Passingham, 1975, 1978) as a measure of "extra capacity" for information processing or integrative functioning, over and above that needed for simple analysis and control of brain-body communication. Comparing analyses presented here of the scaling of brain weight *vs.* each of body weight and spinal cord weight, within mammals the two methods produce virtually identical pictures of the distribution of species according to their relative brain sizes. This is not surprising because spinal cord weight is highly correlated with body weight for this class. However, the relationship between spinal cord weight and body weight is somewhat different for birds and amphibians, and, therefore, analysing brain weight against spinal cord weight gives a rather different impression of the relative brain sizes of mammals, birds and amphibians than analyses against body weight. The greater degree of overlap between the classes when spinal cord weight is used as the baseline suggests that in birds and some mammals at least, brain size may be very similar in relation to the level of nervous communication between the brain and body; and from this point of view, brain development in amphibians may not be so different from the other two classes as studies of brain weight relative to body weight indicate. [The two approaches for the analysis of relative brain size used here, correspond to those categorized by Smith (1992) as "body size allometry" and "biomechanical allometry".]

These results suggest that in broad terms brain size responds to two sets of influences during evolution. Much of the difference between extant classes in brain size relative to body size may be the result of adaptations of the whole CNS to a level of information flow between the brain and body, which is more or less constant within a class. (This could be related to the level of locomotor sophistication and common factors of the mode of locomotion in each class.) Hence, variation between classes in brain size and in spinal cord size is to a considerable extent correlated. Remaining interclass variation in the brain-cord relationship may largely be associated with differences in the inputs from cranial sensory organs, resulting from the major

differences in sensory perception emphasized by Jerison (1973). The “extra capacity” of the brain for analysis and integration of information may vary within a class, and this is reflected in variation in brain size independent of variation in spinal cord size. Living primates, for example, are typical extant mammals in terms of the level of nervous inputs and outputs from brain to body (cord size relative to body size), but the haplorhines have expanded their brain size above that needed to cope with basic analysis and control of these communications (brain size relative to cord size).

Analyses presented here are entirely based on extant species. There is some evidence from primates that spinal cord size has increased during evolution within a class (MacLarnon, in press). This increase was correlated with increasing brain size, and increase in both may have been associated with increasing locomotor sophistication (Martin, 1980). Clearly, further analyses and, particularly, more data on the dimensions of the spinal cord are needed to test the suggested model, but combined analysis of brain size, spinal cord size and body size may be important to understanding the evolutionary basis of brain size changes.

Conclusion

Evidence presented here indicates that the gross dimensions of the spinal cord, its weight and length, vary little relative to body size among primates and other mammals. It has often been suggested that locomotor variation has a strong influence on spinal cord dimensions. On the basis of present results, however, it does not seem to have a major or widespread effect on gross spinal cord size among mammals. For example, despite the very wide range of tail length and usage among primates, available evidence indicates only that completely tailless species have slightly short cords relative to body size.

The limited variation in relative mammalian cord dimensions contrasts greatly with the considerable variation in relative brain size, and the two parts of the central nervous system must have responded to different selection forces within the class. Birds have on average slightly smaller spinal cords and brains than the mammals, and limited evidence suggests that amphibians may have much smaller spinal cords, as well as much smaller brains, than both other classes.

It is apparently the total weight of the spinal cord that scales to body weight as a surface area to a volume, and not its cross-sectional area, unlike previous suggestions. Somatic innervation may, therefore, reflect body surface area in some way, but interpretation is not easy. Present evidence suggests that spinal cord weight scales to body weight with a significantly different exponent from brain weight, and, therefore, its scaling relationship cannot be accounted for by metabolic explanations developed for the other part of the CNS.

Because cord weight is highly correlated with body weight within the mammals, it could be used as a measure of body size in studies of relative mammalian brain size, though it must be remembered that there is some residual variation in relative cord weight. In addition, allometric residuals must be used to represent relative brain size rather than brain:cord ratios if size-related variation is to be removed as effectively as possible, because cord weight and brain weight scale to body weight with different exponents. Between classes, as cord size varies quite markedly relative to body size, it cannot be used to produce size-free indices of relative brain size. However, allometric analysis of brain size relative to cord size provides the basis for meaningful comparison of relative brain size or relative intelligence both within and between classes, because it assesses brain size relative to the level of nervous inputs and outputs to and from the brain and body. This level of information flow may vary with major differences in

locomotor mode and the sophistication of its control, for example between different classes. Brain size may also expand above the level required for brain–cord communication, enabling information processing and integration beyond basic requirements. Extant mammals have large relative brain and cord sizes compared with other classes, perhaps to enable sophisticated control of their complex locomotor patterns. Some groups, for example, the haplorhine primates, have expanded the brain beyond this level for additional cognitive functions.

Acknowledgements

Much of the work presented was carried out at University College London. I am very grateful to Bob Martin for his enthusiasm and advice, and to all my colleagues in the Anthropology Department for their encouragement. Help with the collection of spinal cord data was given by many people, particularly Alan Ainsworth and Mary Brancker. This paper was written while I was on sabbatical leave at the Natural History Museum, London, and I am grateful for the facilities provided there. The manuscript was greatly improved thanks to comments from Leslie Aiello, Harry Jerison and two other anonymous referees.

References

- Aiello, L. C. (1981). The allometry of primate body proportions. *Symp. zool. Soc. Lond.* **48**, 331–358.
- Aiello, L. C. (1992). Size and shape in human evolution. *J. hum. Evol.* **22**, 127–147.
- Aiello, L. C. & Wood, B. A. (1994). Cranial variables as predictors of hominine body mass. *Am. J. phys. Anthropol.* **95**, 409–426.
- Ariëns Kappers, C. U., Huber, G. C. & Crosby, E. C. (1936). *The Comparative Anatomy of the Nervous System of Vertebrates, Including Man*. Vol. 1. New York: Macmillan.
- Armstrong, E. (1982). A look at relative brain size in mammals. *Neurosci. Lett.* **34**, 101–104.
- Cammermeyer, J. (1956) Changes in volume of cat spinal cord induced by formalin fixatives. *J. Neuropath. Exper. Neurol.* **15**, 212–214.
- Dagosto, M. & Terranova, C. J. (1992). Estimating the body size of Eocene primates: a comparison of results from dental and postcranial variables. *Int. J. Primatol.* **13**, 307–344.
- Deacon, T. W. (1990a). Problems of ontogeny and phylogeny in brain-size evolution. *Int. J. Primatol.* **11**, 237–281.
- Deacon, T. W. (1990b). Confusing size-correlated differences with phylogenetic “progression” in brain evolution. *Behav. Brain Sci.* **13**, 185–187.
- Donaldson, H. H. & Davis, D. J. (1903). A description of charts showing the areas of the cross-sections of the human spinal cord at the level of each spinal nerve. *J. Comp. Neurol.* **13**, 19–40.
- Donaldson, H. H. & Hatai, S. (1911). A comparison of the Norway rat with the albino rat in respect to body length, brain weight, spinal cord weight and the percentage of water in both the brain and spinal cord. *J. Comp. Neurol.* **21**, 417–458.
- Fox, J. H. & Wilczynski, W. (1986). Allometry of major CNS division: towards a reevaluation of somatic brain-body scaling. *Brain Behav. Evol.* **28**, 157–169.
- Hofman, M. A. (1982) Encephalization in mammals in relation to the size of the cerebral cortex. *Brain Behav. Evol.* **20**, 84–96.
- Hopf, A. & Claussen, C.-P. (1971). Comparative studies on the fresh weights of the brains and spinal cords of *Theropithecus gelada*, *Papio hamadryas* and *Cercopithecus aethiops*. *Proc. 3rd Int. Congr. Primatol., Zurich 1970*, Vol. 1, pp. 115–121. Basel: Karger.
- Jerison, H. J. (1973). *Evolution of the Brain and Intelligence*. New York: Academic Press.
- Krompecher, St. & Lipák, J. (1966). A simple method for determining cerebralization, brain weight and intelligence. *J. Comp. Neurol.* **127**, 113–120.
- Latimer, H. B. (1950). The weights of the brain and of its parts, and the weight and length of the spinal cord in the adult male guinea pig. *J. Comp. Neurol.* **93**, 37–51.
- Latimer, H. B. & Sawin, P. B. (1955). Morphogenetic studies of the rabbit. XIII. The influence of the dwarf gene upon organ size and variability in race X. *Anat. Rec.* **123**, 447–466.
- Latimer, H. B. & Sawin, P. B. (1957). Morphogenetic studies of the rabbit. XIX. Organ size in relation to body size in large race III and in small race X. *Anat. Rec.* **129**, 457–472.
- MacLarnon, A. M. (1987). Size relationships of the spinal cord and associated skeleton in primates. Ph.D. Dissertation, University of London.

- MacLarnon, A. M. (1993). The vertebral canal. In (A. Walker & R. Leakey, Eds) *The Nariokotome Homo erectus Skeleton*, pp. 359–390. Cambridge, MA: Harvard University Press.
- MacLarnon, A. M. (in press). The evolution of the spinal cord in primates: evidence from the foramen magnum and the vertebral canal. *J. hum. Evol.*
- Martin, R. (1928). *Lehrbuch der Anthropologie*. Vol. 2. Jena: Gustav Fischer.
- Martin, R. D. (1980). Adaptation and body size in primates. *Z. Morph. Anthropol.* **71**, 115–124.
- Martin, R. D. (1981). Relative brain size and basal metabolic rate in terrestrial vertebrates. *Nature* **293**, 57–60.
- Martin, R. D. (1982). Allometric approaches to the evolution of the primate nervous system. In (E. Armstrong & D. Falk, Eds) *Primate Brain Evolution: Methods and Concepts*, pp. 39–56. New York: Plenum.
- Martin, R. D. & Barbour, A. D. (1989). Aspects of line-fitting in bivariate allometric analyses. *Folia primatol.* **53**, 65–81.
- Mehler, W. R. (1969). Some neurological species differences—a *posteriori*. *Ann. N.Y. Acad. Sci.* **167**, 424–468.
- Nayak, U. V. (1933). A comparative study of the Lorinae and Galaginae. Ph.D. Dissertation, University of London.
- Nieuwenhuys, R. (1964). Comparative anatomy of the spinal cord. *Progr. Brain Res.* **11**, 1–57.
- Pagel, M. D. & Harvey, P. H. (1988). Recent developments in the analysis of comparative data. *Q. Rev. Biol.* **63**, 413–440.
- Passingham, R. E. (1975). The brain and intelligence. *Brain Behav. Evol.* **11**, 1–15.
- Passingham, R. E. (1978). Brain size and intelligence in primates. In (D. J. Chivers & K. A. Joysey, Eds) *Recent Advances in Primatology*, Vol. 3, pp. 85–86. London: Academic Press.
- Ravenel, M. (1877). Die Maasverhältnisse der Wirbelsäule und des Rückenmarkes beim Mensch. *Z. Anat. Entwickl.* **II**, 334–356.
- Ridgway, S. H., Flanigan, N. J. & McCormick, J. G. (1966). Brain-spinal cord ratios in porpoises: possible correlations with intelligence and ecology. *Psychon. Sci.* **6**, 491–492.
- Ringo, J. L. (1991). Neuronal interconnection as a function of brain size. *Brain Behav. Evol.* **38**, 1–6.
- Romanes, G. J. (Ed) (1972). *Cunningham's Textbook of Anatomy*, 11th edn. Oxford: Oxford University Press.
- Sarnat, H. B. & Netsky, M. G. (1981). *Evolution of the Nervous System*. 2nd edn. New York: Oxford University Press.
- Smith, R. J. (1992). Categories of allometry: body size versus biomechanics. *J. hum. Evol.* **24**, 173–182.
- Smith, R. J. (1994). Regression models for prediction equations. *J. hum. Evol.* **26**, 239–244.
- Towe, A. L. (1973). Relative numbers of pyramidal tract neurons in mammals of different sizes. *Brain Behav. Evol.* **7**, 1–17.
- Warden, C. J. (1951). Animal intelligence. *Sci. Am.* **184**, 64–68.